

FIND SQUARE APPLET

Java AWT Input & Action Event Handling

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Technology: Java Applet, AWT, Event Listeners

What the Question is Asking

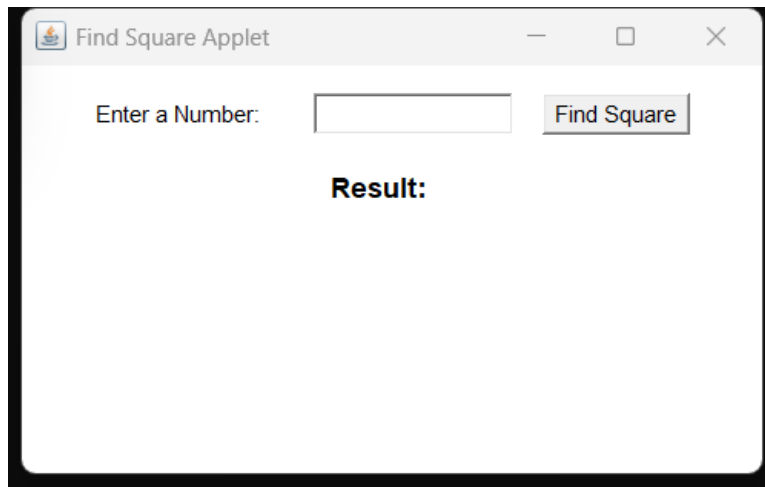
Design an Applet that accepts a number from the user and displays its square when a button is clicked. Use AWT controls and appropriate event handling.

Requirements Covered

Requirement	Implementation Details
AWT Input Controls	Employs Label, TextField for input entry, and Button to execute calculations.
Event Handling	Implements ActionListener interface and registers it to the Button control.
Action Implementation	Extracts text from TextField, parses it as double, computes value square, and sets result label text.
Validation & Error Handlers	catches NumberFormatException to gracefully report non-numeric inputs without crashing.

Actual GUI Output

Below is the screenshot of the active GUI application window taken directly during execution:



Layout Manager Evaluation

Layout	Use	Evaluation
FlowLayout	Applet Layout	Aligns labels, text fields, and calculation buttons sequentially with proper spacing.

Java Source Code

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

/*
<applet code="FindSquareApplet" width="400" height="250">
</applet>
*/
public class FindSquareApplet extends Applet implements ActionListener {
    private TextField inputField;
    private Button calcButton;
    private Label resultLabel;

    @Override
    public void init() {
        setLayout(new FlowLayout(FlowLayout.CENTER, 15, 15));

        add(new Label("Enter a Number:"));
        inputField = new TextField(10);
        add(inputField);

        calcButton = new Button("Find Square");
        add(calcButton);

        resultLabel = new Label("Result: ");
        resultLabel.setFont(new Font("Arial", Font.BOLD, 14));
        add(resultLabel);

        calcButton.addActionListener(this);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        try {
            double num = Double.parseDouble(inputField.getText().trim());
            double square = num * num;
            resultLabel.setText("Result: Square of " + num + " is " + square);
        } catch (NumberFormatException ex) {
            resultLabel.setText("Result: Invalid Input!");
        }
    }

    // Standalone runner
    public static void main(String[] args) {
        Frame frame = new Frame("Find Square Applet");
        FindSquareApplet applet = new FindSquareApplet();
        applet.init();
        applet.start();
        frame.add(applet);
        frame.setSize(400, 250);
        frame.setLocationRelativeTo(null);
        frame.addWindowListener(new WindowAdapter() {
            @Override
            public void windowClosing(WindowEvent e) {
                applet.stop();
                applet.destroy();
                System.exit(0);
            }
        });
        frame.setVisible(true);
    }
}
```

}

How to Run

1. Save code as FindSquareApplet.java
2. Compile using javac: `javac FindSquareApplet.java`
3. Run using java: `java FindSquareApplet`

Conclusion

The Find Square Applet successfully demonstrates mathematical event-driven programming. AWT controls and FlowLayout deliver a clean layout.